BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 3 Packet.

Muscle · Cardiovascular · Blood

BIO 004 · MODULE 3

Contents

Seven chapters, in the order the module teaches them. Part A of the Module 3 packet. Part B is the pre-work, Part C is the lab structure list.

Everything else for Module 3

Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

WHAT MODULE 3 COVERS

Module 3 covers muscle, the cardiovascular system, blood, and the respiratory system. It runs from the sarcomere out to the named muscles of the trunk and upper limb, then through the heart and the vessels that leave it, and ends at the alveolus.

The seven chapters build in order. Muscle tissue is the material, and fascicle arrangement is how that material is packaged into an organ. The heart is a muscle that never rests, so it follows the muscle chapters rather than starting a new topic. The vessels leave the heart, blood is what travels in them.

The named muscles of the back, thorax, and upper limb, and the named arteries, veins, and nerves of the thorax and upper limb, are anatomy lab material. You will not find them in Part A. They are in Part C, and the origin, insertion, action, and innervation of every muscle is on the muscle chart on the Loops page.

This is anatomy. Where function appears in these notes it is marked as a physiology note, and it is there because the structure does not make sense without it.

PART A

Course Notes.

Seven chapters, in the order the module teaches them. Read a chapter before the class day it belongs to, then do the matching pre-work packet in Part B. This is your reference for every video, lab, and exam question in Module 3.

CHAPTER 1

Muscle Structure & Sarcomeres

MUSCLE STRUCTURE AND SARCOMERES

A reference for the muscle structure video and lab. This chapter covers the structure of a skeletal muscle from its connective tissue coverings, down through the muscle fiber, to the sarcomere and its myofilaments. The contraction mechanism belongs to physiology and is not covered here.

BY THE END

1. Name the connective tissue coverings of a skeletal muscle and trace how they connect to the tendon.
2. Order the levels of muscle organization, from the whole muscle down to the myofilament.
3. Identify the parts of a muscle fiber, including the T tubules, sarcoplasmic reticulum, and triad.
4. Diagram a sarcomere, name its bands, lines, and zones, and state which filaments lie in each.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Muscle structure

drsrennie-stack.github.io/new-build-bio4-solano/muscle-structure-concept-videos.html

Skeletal Muscle, an Overview

A skeletal muscle is an organ. It is built of muscle tissue plus connective tissue, blood vessels, and nerves, and it is wrapped, bundled, and anchored at every level of its structure.

THE MUSCLE AS AN ORGAN

Structure	What it is
Skeletal muscle	A voluntary, striated muscle organ, usually attached to bone, that produces movement when it contracts
Muscle fiber	A single skeletal muscle cell, long and cylindrical, with many nuclei
Origin	The attachment of a muscle to the more stationary bone
Insertion	The attachment of a muscle to the more movable bone, the bone that is pulled toward the origin
Belly	The thick, fleshy middle region of the muscle between its attachments

Attachments and Action

A muscle crosses at least one joint and attaches to the articulating bones. When it contracts, one bone stays put and the other moves.

ORIGIN, INSERTION, ACTION

Structure	What it is
Origin	The tendon's attachment to the stationary bone, usually the proximal one
Insertion	The tendon's attachment to the movable bone, usually the distal one, pulled toward the origin
Action	The movement the muscle produces when it contracts

How Muscles Work Together

Skeletal muscles are arranged in opposing pairs across a joint, and each contraction has muscles playing different roles.

THE ROLES IN A MOVEMENT

Structure	What it is
Agonist (prime mover)	The muscle that contracts to produce the action
Antagonist	The opposing muscle that stretches and yields to the prime mover
Synergist	A muscle that aids the action or prevents unwanted movement
Fixator	A muscle that stabilizes the origin so the prime mover acts efficiently
Compartment	A group of muscles, vessels, and nerves with a common function, such as the flexor and extensor compartments

How Skeletal Muscles Are Named

A muscle's name usually encodes one or more of these features. Learn the clues and the name tells you the muscle.

NAMING CLUES AND WHAT THEY LOOK LIKE IN A NAME

Naming clue	Examples in the name
Direction of fibers	rectus, transverse, oblique
Size	maximus, minimus, longus, brevis, vastus
Shape	deltoid, trapezius, serratus, rhomboid, orbicularis
Action	flexor, extensor, abductor, adductor, levator, pronator, supinator
Number of origins	biceps, triceps, quadriceps
Location	temporalis, over the temporal bone
Origin and insertion	sternocleidomastoid, sternum and clavicle to the mastoid process

The Three Muscle Tissue Types

All three are muscle, but their structure differs. Compare them by location, appearance, control, and cell features.

LOCATION, CONTROL, AND THE CELL

Type	Location	Striations and control	Cell features
Skeletal	Attached to the skeleton	Striated, voluntary	Long cylindrical fibers, many nuclei pushed to the edge
Cardiac	The heart wall	Striated, involuntary	Branching fibers, one central nucleus, joined by intercalated discs with desmosomes for adhesion and gap junctions for electrical coupling
Smooth	Walls of hollow organs and vessels	Non-striated, involuntary	Spindle-shaped cells, one central nucleus

Connective Tissue Coverings

Three connective tissue sheaths wrap the muscle at three scales. They are continuous with each other and with the tendon, so a fiber's pull is carried all the way to the bone.

THE THREE SHEATHS

Sheath	What it wraps	Tissue
Epimysium	The entire muscle, the outermost layer	Dense irregular connective tissue
Perimysium	Each fascicle, a bundle of muscle fibers	Dense irregular connective tissue
Endomysium	Each individual muscle fiber	A fine sheath of areolar connective tissue with reticular fibers

FASCIA AND THE ATTACHMENTS THEY FORM

Structure	What it is
Superficial fascia	The subcutaneous layer of areolar and adipose tissue between the skin and the muscles
Deep fascia	A sheet of dense connective tissue that wraps muscles, separates them, and binds those with a common action into functional groups
Tendon	A cord of dense regular connective tissue, the merged continuation of all three sheaths, that attaches a muscle to bone
Aponeurosis	A broad, flat sheet of tendon-like connective tissue, an attachment used where a muscle is wide

Levels of Organization

A skeletal muscle is a set of structures nested inside one another. Follow them in order, from the whole organ down to the protein strands.

1. **Muscle** the whole organ, wrapped in epimysium
2. **Fascicle** a visible bundle of muscle fibers, wrapped in perimysium
3. **Muscle fiber** a single muscle cell, wrapped in endomysium
4. **Myofibril** a long contractile rod running the length of the fiber; hundreds to thousands fill each fiber
5. **Myofilament** the thick and thin protein strands within a myofibril; their ordered overlap forms the sarcomere

The Muscle Fiber

The muscle fiber is the muscle cell. Its parts carry the names of ordinary cell structures with the prefix sarco, from the Greek for flesh.

INSIDE THE FIBER

Structure	What it is
Sarcolemma	The plasma membrane of the muscle fiber
Sarcoplasm	The cytoplasm of the muscle fiber
Myonuclei	The many nuclei of the fiber, flattened against the inner sarcolemma; a muscle fiber is multinucleated
Myofibrils	The contractile rods that fill most of the sarcoplasm
Transverse (T) tubules	Narrow tunnels of the sarcolemma that dive inward, carrying the surface signal deep into the fiber
Sarcoplasmic reticulum	The smooth endoplasmic reticulum of the muscle fiber, a network of tubules wrapping each myofibril; it stores calcium ions
Terminal cisternae	The enlarged end sacs of the sarcoplasmic reticulum, set on either side of a T tubule
Triad	The structural unit of one T tubule flanked by two terminal cisternae
Mitochondria	Packed in rows between the myofibrils, they supply the ATP that contraction demands
Myoglobin	A red, oxygen-binding pigment in the sarcoplasm that stores oxygen and gives muscle its color
Glycogen	Granules of stored carbohydrate fuel held in the sarcoplasm

Myofilaments, Thick and Thin

Two kinds of protein filament fill each myofibril. Compare them by their main protein and structure.

THE TWO MYOFILAMENTS

Myofilament	Main protein	Structure	Anchored at
Thick filament	Myosin	A bundle of myosin molecules; each molecule has a long tail and a pair of heads, and the heads project outward as cross bridges	The M line
Thin filament	Actin	Two twisted strands of actin subunits; each subunit carries a myosin-binding site	The Z disc

THE TWO REGULATORY PROTEINS

Structure	What it is
Tropomyosin	A rod-shaped regulatory protein that lies end to end along the thin filament
Troponin	A three-part regulatory protein bound at intervals to tropomyosin on the thin filament

The Sarcomere

The sarcomere is the contractile unit of muscle, the segment of a myofibril from one Z disc to the next. Its bands and zones are named for how the overlapping filaments look under the microscope.

COMPARE THE REGIONS

Region	What it is	Filaments present
Z disc	The protein plate at each end of the sarcomere, its border	Anchors the thin filaments
A band	The dark band, as wide as the thick filaments are long	Thick filaments across its full width, with thin filaments overlapping at each edge
I band	The light band; a Z disc runs through its center, so one I band is shared by two sarcomeres	Thin filaments only
H zone	The lighter region in the center of the A band	Thick filaments only
M line	The protein line in the center of the H zone	Anchors the thick filaments to one another
Zone of overlap	The region at each end of the A band	Both thick and thin filaments, interleaved

WHAT YOU SEE ON THE SLIDE

Structure	What it is
Striations	The stripes of skeletal muscle, the alternating dark A bands and light I bands of thousands of aligned sarcomeres

Structural Proteins of the Sarcomere

Besides the contractile proteins, a set of structural proteins holds the sarcomere in register and links it to the cell membrane. Compare them by location and role.

LOCATION AND ROLE

Protein	Location	Role
Titin	Spans from the Z disc to the M line	Anchors the thick filament, gives the sarcomere elastic recoil, and keeps it from overstretching
Nebulin	Runs the length of the thin filament	Anchors the thin filament and sets its length
Alpha-actinin	The Z disc	Binds the thin filaments and titin to the Z disc
Myomesin	The M line	Binds the thick filaments together at the M line
Dystrophin	Links the thin filaments to the sarcolemma	Transfers the force of contraction from the sarcomere out to the connective tissue; faulty in muscular dystrophy

HOLD ONTO THIS

The A band never changes length, because it is the thick filament. The I band and the H zone are the ones that narrow. Any question about band width comes back to that.

STRUCTURE EXPLAINS THE INJURY

The three connective tissue sheaths are continuous with one another and with the **tendon**, so the pull of every fiber reaches the bone. This continuity also explains injury. A muscle strain tears fibers and endomysium within the belly, while a tendon rupture separates the muscle from the bone entirely, and the two heal on very different timelines.

Muscular dystrophy is a group of inherited diseases of muscle structure. In Duchenne muscular dystrophy the gene for **dystrophin** is faulty, so the link between the sarcomere and the sarcolemma fails, the fiber damages itself with ordinary use, and weakness is progressive.

The **triad**, one T tubule flanked by two terminal cisternae, is the structural meeting point where a surface signal reaches the calcium store of the sarcoplasmic reticulum. Its tight, repeating arrangement is what lets an entire fiber respond almost at once.

CHAPTER 2

Fascicle Arrangement & Lever Systems

MUSCLE ARRANGEMENT OF FASCICLES AND LEVER SYSTEMS

A reference for the fascicle arrangement and lever systems video and lab. This chapter covers how the fascicles of a muscle are arranged, how muscles are named, how they work in groups, and the bone-and-joint lever systems they act on.

BY THE END

1. Identify the patterns of fascicle arrangement and give an example muscle for each.
2. Explain how fascicle architecture trades range of motion against power.
3. Name the criteria used to name muscles, and classify the roles muscles play in a movement.
4. Identify the parts of a lever and classify first, second, and third class levers in the body.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Fascicle arrangement and levers

drsrennie-stack.github.io/new-build-bio4-solano/fascicle-lever-concept-videos.html

From Fascicles to Movement, an Overview

A skeletal muscle moves a bone by pulling across a joint. How its fascicles are arranged, what it is named, the team it works in, and the lever it acts on all shape the movement it produces.

THE STARTING VOCABULARY

Structure	What it is
Fascicle	A bundle of muscle fibers wrapped in perimysium, the unit whose arrangement gives a muscle its shape
Origin	The attachment of a muscle to the more stationary bone
Insertion	The attachment of a muscle to the more movable bone, pulled toward the origin
Action	The movement a muscle produces when it contracts and pulls the insertion toward the origin

Fascicle Arrangement Patterns

The fascicles of a muscle run in a set pattern, and that pattern gives the muscle its overall shape. Compare the patterns by how the fascicles lie and an example muscle.

PATTERN, ARRANGEMENT, AND EXAMPLE

Pattern	Fascicle arrangement	Example muscle
Parallel	Fascicles run parallel to the long axis of the muscle	Sartorius, rectus abdominis
Fusiform	Fascicles run parallel, but the muscle has a thick central belly that tapers to a tendon at each end	Biceps brachii
Circular	Fascicles arranged in concentric rings around an opening, forming a sphincter	Orbicularis oris, orbicularis oculi
Convergent	A broad origin whose fascicles converge onto a single tendon, giving a triangular or fan shape	Pectoralis major
Pennate	Short fascicles attach at an angle to a central tendon, like the barbs of a feather. Unipennate has fascicles on one side, bipennate on both, multipennate on a branched tendon	Unipennate, extensor digitorum; bipennate, rectus femoris; multipennate, deltoid

Architecture, Power, and Range

Fascicle architecture is a trade-off. A muscle can be built for a wide range of motion or for power, but not for both at once. Compare the two extremes.

THE TRADE-OFF

Architecture	Fascicle length and number	Range of motion	Power
Parallel and fusiform	Long fascicles, fewer of them per unit of muscle	Larger range of motion, the muscle shortens by a greater distance	Less power, fewer fibers pull on the tendon
Pennate	Short fascicles, many of them packed at an angle per unit of muscle	Smaller range of motion, the angled fascicles shorten by a shorter distance	Greater power, many more fibers pull on the tendon

WHY THE TRADE-OFF EXISTS

Structure	What it is
Range of shortening	A muscle fiber can shorten to roughly two thirds of its resting length, so a muscle with longer fascicles moves its insertion farther

How Muscles Are Named

A muscle's name is a description. Most names combine two or more of these criteria, so the name itself tells you where the muscle is, what it looks like, or what it does.

THE NAMING CRITERIA

Naming criterion	What it describes	Examples
Direction of fascicles	How the fascicles run relative to the body's midline	rectus, straight; oblique, at an angle; transversus, across
Relative size	The size of the muscle compared with its neighbors	maximus, minimus, longus, brevis
Shape	The overall geometric shape of the muscle	deltoid, triangular; trapezius, trapezoid; rhomboid
Location	The bone or body region the muscle is near	tibialis anterior, frontalis, intercostals
Number of origins	How many heads, or origin tendons, the muscle has	biceps, two; triceps, three; quadriceps, four
Origin and insertion	The muscle's two attachment points	sternocleidomastoid, from the sternum and clavicle to the mastoid process
Action	The movement the muscle produces	flexor, extensor, abductor, adductor, levator

Muscles Working in Groups

Almost no movement is the work of one muscle. Muscles act in teams, and the same muscle can play different roles in different movements.

THE FOUR ROLES

Role	Function in the movement	Example
Agonist	The prime mover, the muscle chiefly responsible for producing a given movement	Biceps brachii during elbow flexion
Antagonist	Opposes the agonist, relaxing and lengthening as the agonist contracts; it can slow or reverse the movement	Triceps brachii during elbow flexion
Synergist	Assists the agonist, adding force or preventing an unwanted movement	Brachialis assisting elbow flexion
Fixator	A synergist that holds the origin bone steady so the agonist can act efficiently on the insertion bone	Scapular muscles steadying the scapula during arm movements

Lever Systems

When a muscle pulls on a bone, the bone acts as a lever and the joint as its pivot. Every body movement is a lever at work.

THE PARTS OF A LEVER

Structure	What it is
Lever	A rigid bar that moves around a fixed point; in the body, a bone is the lever
Fulcrum	The fixed point the lever turns around; in the body, a joint is the fulcrum
Effort	The force applied to move the lever; in the body, the pull of a contracting muscle
Load	The resistance moved by the lever; in the body, the weight of the body part plus anything it carries
Mechanical advantage	When the effort acts farther from the fulcrum than the load, so a small effort moves a large load
Mechanical disadvantage	When the effort acts closer to the fulcrum than the load, so a large effort is needed, but the load moves fast and far

The three classes of lever

Levers are sorted into three classes by the order of the fulcrum, effort, and load along the bar.

ARRANGEMENT AND AN EXAMPLE IN THE BODY

Lever class	Arrangement along the lever	Example in the body
First class lever	The fulcrum lies between the effort and the load; think of a seesaw	Nodding the head: the posterior neck muscles, the atlanto-occipital joint, and the weight of the face
Second class lever	The load lies between the fulcrum and the effort; think of a wheelbarrow	Standing on tiptoe: the ball of the foot, the body's weight, and the calf muscles pulling up the heel
Third class lever	The effort lies between the fulcrum and the load; think of tweezers. The most common lever in the body	Flexing the elbow: the elbow joint, the biceps pull, and the weight held in the hand

HOLD ONTO THIS

Most body movements run on third class levers, so most muscles work at a mechanical disadvantage. The body trades force for speed and range, which is why a muscle has to pull far harder than the weight it is lifting.

ARCHITECTURE READS OFF THE BODY

Fascicle architecture is visible in the clinic. A **pennate** muscle like the deltoid packs many short fascicles around a tendon for power, while a **parallel** muscle like the sartorius trades power for a long, sweeping range. A surgeon planning a tendon transfer matches the architecture to the job the muscle must take over.

Muscle names are a working vocabulary. Reading flexor carpi radialis tells a clinician the action (flexor), the region (carpi, the wrist), and the position (radialis, the radial side) before the muscle is ever seen, which is why the naming criteria are worth learning as a system.

Most body movements run on **third class levers**, which work at a mechanical disadvantage: the muscle must generate a large force, but it buys speed and a wide range of motion. That trade-off is why a small change in where a tendon inserts can noticeably change a joint's strength and speed.

CHAPTER 5

The Heart

BLOCK 3 · MODULE 2: STRUCTURES, VALVES, AND THE PATHWAY OF BLOOD

A reference for the heart video and lab. This chapter covers the heart wall and pericardium, the four chambers and four valves, the fibrous skeleton, cardiac muscle tissue, the pathway of blood, and the great vessels and coronary circulation. The focus is on structure.

BY THE END

1. Name the layers of the heart wall and the pericardium.
2. Identify the four chambers and the four valves, and what each one separates.
3. Identify the internal features of each chamber.
4. Describe cardiac muscle tissue and the intercalated disc.
5. Trace the pathway of blood through the heart in order.
6. Name the great vessels and the coronary arteries and veins.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The heart

drsrennie-stack.github.io/new-build-bio4-solano/heart-concept-videos.html

The Heart, an Overview

The heart is a hollow, muscular pump about the size of a closed fist. It sits in the mediastinum and drives blood through two circuits, one to the lungs and one to the body.

LOCATION AND ORIENTATION

Structure	What it is
Heart	A hollow, muscular organ that pumps blood through the blood vessels
Location	In the mediastinum, between the two lungs, resting on the diaphragm and tilted to the left
Mass	About two thirds of the heart's mass lies to the left of the midline
Apex	The blunt, rounded inferior tip, pointing down, forward, and to the left. It is the tip of the left ventricle
Base	The broad superior surface, mostly left atrium, where the great vessels enter and leave
Anterior surface	The sternocostal surface, mostly right ventricle, lying under the sternum and ribs
Inferior surface	The diaphragmatic surface, mostly left ventricle, resting on the diaphragm
Pulmonary circuit	The loop the right side of the heart drives, carrying blood to the lungs and back
Systemic circuit	The loop the left side of the heart drives, carrying blood to the body and back

The Heart Wall and Pericardium

The heart is wrapped in a double sac, the pericardium, and its own wall is built in three layers.

THE PERICARDIUM

Structure	What it is
Pericardium	The double-walled sac that encloses the heart
Fibrous pericardium	The tough, inelastic outer layer of dense connective tissue. It anchors the heart in the mediastinum and prevents overfilling
Serous pericardium	The inner double membrane: a parietal layer lining the fibrous pericardium and a visceral layer on the heart surface
Pericardial cavity	The space between the two serous layers, holding a film of serous fluid that reduces friction as the heart beats

The three layers of the heart wall

COMPARE THEM FROM OUTSIDE IN

Layer	Position	Description
Epicardium	The outer layer	The visceral layer of the serous pericardium. A thin serous membrane on the heart surface, with fat and the coronary vessels running in it
Myocardium	The thick middle layer	Cardiac muscle tissue in spiralling bundles. About ninety five percent of the wall thickness, and the layer that contracts
Endocardium	The inner layer	A smooth endothelium lining the chambers and covering the valves, continuous with the lining of the great vessels

HOLD ONTO THIS

The epicardium and the visceral serous pericardium are the same membrane under two names. Which name you use depends on whether you are describing the heart wall or the sac.

The Chambers

The heart has four chambers: two thin-walled atria that receive blood and two thick-walled ventricles that pump it out.

WHAT EACH CHAMBER RECEIVES AND WHERE IT SENDS BLOOD

Chamber	Receives blood from	Pumps blood to
Right atrium	The body, through the superior and inferior venae cavae, and the heart wall through the coronary sinus	The right ventricle
Right ventricle	The right atrium	The lungs, through the pulmonary trunk
Left atrium	The lungs, through the four pulmonary veins	The left ventricle
Left ventricle	The left atrium	The body, through the aorta. Its wall is the thickest of the four chambers

Internal features

THE NAMED STRUCTURES INSIDE THE CHAMBERS

Structure	What it is
Interatrial septum	The wall between the two atria
Fossa ovalis	The oval depression in the interatrial septum, the remnant of the fetal foramen ovale
Interventricular septum	The thick wall between the two ventricles
Auricle	A small wrinkled pouch on each atrium that slightly increases its capacity
Pectinate muscles	Parallel muscular ridges on the walls of the atria and the auricles
Crista terminalis	The ridge in the right atrium separating the smooth posterior wall from the rough pectinate anterior wall
Sinus venarum	The smooth-walled posterior part of the right atrium, where the venae cavae empty
Trabeculae carneae	The irregular muscular ridges on the inner walls of the ventricles
Papillary muscles	Cone-shaped muscular projections of the ventricle wall that anchor the chordae tendineae
Moderator band	A muscular band in the right ventricle, also called the septomarginal trabecula, that carries part of the conduction system to the anterior papillary muscle
Conus arteriosus	The smooth outflow tract of the right ventricle, leading to the pulmonary valve
Aortic vestibule	The smooth outflow tract of the left ventricle, leading to the aortic valve

The Heart Valves

Four valves keep blood moving in one direction. Each opens to let blood pass and closes to stop it flowing back.

COMPARE THEM BY TYPE AND LOCATION

Valve	Type	Location	Prevents backflow into
Tricuspid valve	Atrioventricular, three cusps	Between the right atrium and the right ventricle	The right atrium
Mitral valve	Atrioventricular, two cusps; also called the bicuspid valve	Between the left atrium and the left ventricle	The left atrium
Pulmonary valve	Semilunar, three cusps	Between the right ventricle and the pulmonary trunk	The right ventricle
Aortic valve	Semilunar, three cusps	Between the left ventricle and the aorta	The left ventricle

WHAT HOLDS THE ATRIOVENTRICULAR VALVES SHUT

Structure	What it is
Chordae tendineae	Strong fibrous cords that tie the cusps of an atrioventricular valve to the ventricle wall
Papillary muscles	The muscular projections that anchor the chordae tendineae and hold the valves shut against the pressure of a contracting ventricle
Fibrous skeleton	The rings of dense connective tissue around the four valve openings. It anchors the valves, gives the cardiac muscle a point of attachment, and electrically insulates the atria from the ventricles

HOLD ONTO THIS

Only the atrioventricular valves have chordae tendineae and papillary muscles. The semilunar valves have neither, because their pocket-shaped cusps fill from above and close themselves. If a question mentions chordae, the answer is tricuspid or mitral.

Cardiac Muscle Tissue

The myocardium is built of cardiac muscle, a tissue found nowhere else in the body. Its structure lets the whole heart contract as one coordinated unit.

CARDIAC MUSCLE AT THE CELLULAR LEVEL

Structure	What it is
Cardiac muscle	The involuntary, striated muscle tissue that forms the myocardium
Cardiomyocyte	A cardiac muscle cell, short and branching, usually with a single central nucleus
Striations	Cardiac muscle is striped, like skeletal muscle, because it too is built of ordered sarcomeres
Intercalated discs	The specialised junctions where neighbouring cardiomyocytes meet end to end
Desmosomes	Anchoring junctions within the intercalated discs that keep the cells from pulling apart
Gap junctions	Channels within the intercalated discs that let the contraction signal pass directly from cell to cell
Abundant mitochondria	Cardiac muscle is packed with mitochondria, about a quarter of the cell volume, to fuel its constant lifelong work

The Pathway of Blood

Follow one drop of blood through the heart, in order. It makes a full loop: into the right side, out to the lungs, back into the left side, and out to the body.

1. **Right atrium** oxygen-poor blood returns from the body through the superior and inferior venae cavae
2. **Tricuspid valve** blood passes from the right atrium down into the right ventricle
3. **Right ventricle** contracts and pumps the blood upward and out
4. **Pulmonary valve** blood passes into the pulmonary trunk, which branches to the lungs
5. **Lungs** blood picks up oxygen and releases carbon dioxide, then returns through the pulmonary veins
6. **Left atrium** receives the now oxygen-rich blood from the pulmonary veins
7. **Mitral valve** blood passes from the left atrium down into the left ventricle

8. **Left ventricle** the thickest chamber contracts and pumps the blood out with force
9. **Aortic valve** blood passes into the aorta and out to the body, completing the loop

The Great Vessels and Coronary Circulation

The great vessels are the large pipes attached to the base of the heart. The heart wall has its own blood supply, the coronary circulation.

THE GREAT VESSELS

Structure	What it is
Superior vena cava	Returns oxygen-poor blood from the head, neck, and upper limbs to the right atrium
Inferior vena cava	Returns oxygen-poor blood from the trunk and lower limbs to the right atrium
Pulmonary trunk	Carries blood from the right ventricle and branches into the right and left pulmonary arteries
Pulmonary veins	Four veins that return oxygen-rich blood from the lungs to the left atrium
Aorta	The largest artery, carrying oxygen-rich blood from the left ventricle to the body

The coronary arteries

The two coronary arteries leave the base of the aorta, run in the grooves on the heart's surface, and branch over the myocardium. The flow is aorta, coronary arteries, capillaries of the myocardium, cardiac veins, coronary sinus, right atrium.

CORONARY ARTERIES AND WHAT EACH SUPPLIES

Coronary artery	Main branches	Region supplied
Right coronary artery (RCA)	Right marginal artery; posterior interventricular artery, the posterior descending or PDA	Right atrium, most of the right ventricle, part of the left ventricle, the posterior septum, and usually the SA and AV nodes
Left coronary artery (LCA)	Anterior interventricular artery (LAD); circumflex artery; left marginal artery	Left atrium, left ventricle, anterior septum, and the apex

The coronary veins

VENOUS DRAINAGE OF THE HEART WALL

Structure	What it is
Coronary sinus	The large vein on the posterior heart that collects the cardiac veins and empties into the right atrium
Great cardiac vein	Runs with the anterior interventricular artery (LAD)
Middle cardiac vein	Runs with the posterior interventricular artery (PDA)
Small cardiac vein	Runs with the right marginal artery
Anterior cardiac veins	Drain the right ventricle directly into the right atrium, bypassing the coronary sinus

HOLD ONTO THIS

Every cardiac vein you can name runs alongside a coronary artery of the matching name. Learn the arteries and the veins come free, because they travel in the same grooves.

STRUCTURE EXPLAINS THE EMERGENCY

The **coronary arteries** are the heart's own supply line, and they are narrow. When one is blocked, the patch of myocardium beyond it is starved of oxygen and its muscle begins to die. That is a heart attack, a myocardial infarction, and it is why the coronary vessels matter far beyond their small size.

A **valve** that does not close cleanly lets blood slip backward, and the turbulence is heard through a stethoscope as a murmur. The **chordae tendineae** and **papillary muscles** are what hold the atrioventricular valves shut against the force of a contracting ventricle; if a papillary muscle is damaged, the valve can fail.

Fluid building up in the **pericardial cavity**, called cardiac tamponade, squeezes the heart from outside and limits how much it can fill. The same fluid film that normally lets the heart glide can, in excess, stop it from working.

CHAPTER 6

The Cardiac Conduction System

BLOCK 3 · MODULE 3: CONDUCTION AND DISORDERS

A reference for the cardiac conduction video and lab. This chapter covers the conduction pathway and its components, the nerve supply to the heart, what an electrocardiogram records, and common conduction disorders.

BY THE END

1. Trace the conduction pathway from the SA node to the Purkinje fibers.
2. Locate each component of the conduction system and state its role.
3. Describe the nerve supply to the heart and what an ECG records.
4. Name common conduction disorders.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Cardiac conduction

drsrennie-stack.github.io/new-build-bio4-solano/conduction-concept-videos.html

The Conduction System, an Overview

The heart sets its own beat. A network of specialised cardiac muscle cells, the conduction system, generates each heartbeat and spreads it so the chambers contract in the right order.

WHAT THE SYSTEM IS

Structure	What it is
Cardiac conduction system	The network of specialised cardiac muscle cells that generates and spreads the signal for each heartbeat. About one percent of the cardiac fibers
Autorhythmicity	The heart's built-in ability to generate its own rhythm without a signal from the nervous system
Conducting cells	Specialised non-contractile cardiac cells that generate and carry the electrical signal
Contractile cells	The ordinary working cardiomyocytes that contract when the signal reaches them
Why the order matters	The conduction system makes the atria contract first and the ventricles second, so each chamber fills before it empties

The Conduction Pathway

Each heartbeat travels a fixed route through the conduction system. Follow the signal in order, from the pacemaker to the ventricle walls.

1. **SA node** the sinoatrial node, the pacemaker; it fires first and sets the pace of the whole heart
2. **Across the atria** the signal spreads through both atria, and the atria contract
3. **AV node** the atrioventricular node; the signal pauses here briefly, giving the atria time to finish emptying
4. **AV bundle** the atrioventricular bundle, or bundle of His; it carries the signal into the interventricular septum, the only electrical link between the atria and the ventricles
5. **Bundle branches** the right and left bundle branches carry the signal down both sides of the interventricular septum toward the apex
6. **Purkinje fibers** spread the signal through the walls of both ventricles, and the ventricles contract from the apex upward

The Components of the Conduction System

COMPARE EACH BY WHERE IT SITS AND WHAT IT DOES

Structure	Location	Role
SA node	The wall of the right atrium, near the opening of the superior vena cava	The natural pacemaker; it fires fastest, about sixty to one hundred times a minute, and sets the heart rate
AV node	The floor of the right atrium, near the interatrial septum	Delays the signal briefly, then passes it on to the ventricles
AV bundle	The superior part of the interventricular septum	The only electrical bridge from the atria to the ventricles
Bundle branches	The right and left sides of the interventricular septum	Carry the signal down toward the apex of the heart
Purkinje fibers	Throughout the walls of both ventricles	Deliver the signal to the contractile cells of the ventricle walls

HOLD ONTO THIS

The fibrous skeleton insulates the atria from the ventricles, so the AV bundle is the only route through. That single bridge is why a block at the AV node or bundle disconnects the two halves of the heartbeat entirely.

Nerve Supply to the Heart

The conduction system runs on its own, but the nervous system can speed it up or slow it down. Autonomic nerves reach the SA and AV nodes and adjust the rate to the body's needs.

THE NERVES THAT MODULATE THE RATE

Structure	What it is
Cardiac centers	Regions of the medulla oblongata in the brainstem that adjust the heart rate
Cardiac plexus	The network of autonomic nerve fibers near the base of the heart through which these nerves reach the nodes
Sympathetic nerves	Autonomic nerves that reach the heart through the cardiac plexus and speed the heart rate
Parasympathetic nerves	Autonomic fibers carried in the vagus nerve that slow the heart rate
A modulating role	These nerves do not start the heartbeat; they only adjust the pace the SA node already sets

HOLD ONTO THIS

A transplanted heart has had its nerves cut and it still beats. That is the cleanest proof that autorhythmicity belongs to the conduction system and not to the nervous system.

The Electrocardiogram

An electrocardiogram, an ECG or EKG, is a tracing of the heart's electrical activity recorded from the body surface. Each deflection of one heartbeat lines up with an event in the conduction system.

COMPARE THE THREE DEFLECTIONS

Deflection	Conduction event	What follows
P wave	The signal spreads across the atria	The atria contract
QRS complex	The signal spreads across the ventricles	The ventricles contract
T wave	The ventricles electrically recover	The ventricles relax

Conduction Disorders

When the conduction system fails, the rhythm goes wrong. Compare the common disorders by what happens and why it matters.

THE COMMON CONDUCTION DISORDERS

Disorder	What happens	Why it matters
Arrhythmia	Any deviation from the normal heart rhythm	The umbrella term for the disorders below
Tachycardia	An abnormally fast resting heart rate	The chambers may not have time to fill between beats
Bradycardia	An abnormally slow resting heart rate	The body may not receive enough blood flow
Heart block	The signal is delayed or stopped at the AV node	The atria and ventricles can beat out of step
Atrial fibrillation	Rapid, disorganised signals in the atria	The atria quiver instead of contracting; common and treatable
Ventricular fibrillation	Rapid, disorganised signals in the ventricles	The ventricles cannot pump; a life-threatening emergency
Ectopic pacemaker	A site outside the SA node takes over the rhythm	The normal pacing order of the heart is lost

WHEN THE RHYTHM NEEDS HELP

An **artificial pacemaker** is a small implanted device that delivers a timed electrical signal to the heart. It takes over the job of a failing **SA node** or carries the signal past a block at the **AV node**, restoring an ordered beat.

In **ventricular fibrillation** the ventricles only quiver, so they move no blood. A defibrillator delivers a shock that stops all the chaotic signals at once, giving the SA node a chance to restart the heart in its normal rhythm.

The brief delay at the **AV node** is not a flaw, it is essential timing. It holds the signal just long enough for the atria to finish emptying into the ventricles before the ventricles contract, and an **ECG** is how a clinician reads whether that timing is intact.

The **SA node** is supplied by the right coronary artery in most people. That is why an occlusion of the RCA can produce a rhythm problem as well as a wall of dying muscle.

CHAPTER 7

Blood Vessels, Structure and Types

BLOCK 3 · MODULE 4: VESSEL ANATOMY AND HISTOLOGY

A reference for the blood vessels video and lab. This chapter covers the three tunics of a vessel wall, the five vessel types, the kinds of artery, capillary, and vein, and the circulatory routes blood follows. Read it with a slide in front of you; almost all of it is visible histology.

BY THE END

1. Name the three tunics of a vessel wall and what each contains.
2. Compare the five vessel types from artery to vein.
3. Distinguish elastic, muscular, and arteriolar arteries on a slide.
4. Distinguish the three kinds of capillary and say where each is found.
5. Describe the circulatory routes, including portal systems and anastomoses.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Blood vessels

drsrennie-stack.github.io/new-build-bio4-solano/vessels-concept-videos.html

Blood Vessels, an Overview

Blood vessels are the body's plumbing. Blood leaves the heart in arteries, reaches the tissues through capillaries, and returns in veins.

THE BASIC VOCABULARY

Structure	What it is
Blood vessel	A tube that carries blood through the body
Artery	A vessel that carries blood away from the heart
Vein	A vessel that carries blood toward the heart
Capillary	The smallest vessel, where exchange between the blood and the tissues takes place
Lumen	The hollow interior of a vessel, the channel the blood flows through
The vascular sequence	Arteries branch into arterioles, then capillaries, which merge into venules, then veins

The Three Tunics

The wall of an artery or a vein is built in three concentric layers, the tunics. A capillary wall is just the innermost layer.

COMPARE THE THREE

Tunic	Position	Tissue	Note
Tunica intima	The innermost layer	A lining of simple squamous endothelium on a basement membrane	The only layer a capillary has. Continuous with the endocardium of the heart
Tunica media	The middle layer	Smooth muscle and elastic fibers	The thickest layer in arteries. Its smooth muscle sets the vessel's diameter
Tunica externa	The outermost layer	Mostly collagen fibers, with nerves and small vessels	Anchors the vessel to the surrounding tissue. The thickest layer in veins

THREE MORE STRUCTURES IN THE WALL

Structure	What it is
Internal elastic lamina	A fenestrated sheet of elastic fibers at the boundary between the tunica intima and the tunica media
External elastic lamina	The elastic sheet separating the tunica media from the tunica externa
Vasa vasorum	The small vessels within the tunica externa that supply the wall of a large vessel. Easily seen on the aorta

HOLD ONTO THIS

The two elastic laminae are the boundaries of the tunica media. If you can find them on a slide you have found the middle layer, and the thickness of what lies between them tells you whether you are looking at an artery or a vein.

The Vessel Types

Blood passes through five kinds of vessel in series, each with a wall built for its job.

COMPARE THEM

Vessel	Direction of flow	Wall	Role
Arteries	Away from the heart	Thick walls with a heavy tunica media	Carry blood under high pressure
Arterioles	Away from the heart, the smallest arteries	Thinner walls, one to two layers of smooth muscle	Adjust how much blood enters a capillary bed. The chief site of resistance
Capillaries	Connect arterioles to venules	A single layer, the tunica intima only	The exchange vessels; gases and nutrients cross here
Venules	Toward the heart, the smallest veins	Thin, porous walls	Collect blood drained from the capillaries; the site where white blood cells leave the bloodstream
Veins	Toward the heart	Thinner walls and a wide lumen, with valves	Carry blood under low pressure

Arteries

Arteries carry blood away from the heart under high pressure. They come in three sizes, and the smallest control how much blood reaches the tissues.

THE THREE SIZES OF ARTERY

Artery	Tunica media	Examples	Function
Elastic artery	Thick, dominated by elastic lamellae; well-defined internal and external elastic laminae	Aorta, pulmonary trunk, brachiocephalic, subclavian, common carotid, common iliac	Stretches with each heartbeat and recoils between beats, propelling blood while the ventricles relax
Muscular artery	Mostly smooth muscle, up to forty layers; thin external elastic lamina	Femoral, axillary, brachial, radial	Distributes blood to specific organs and maintains vascular tone
Arteriole	One or two layers of smooth muscle; internal elastic lamina disappears at the terminal end	Microscopic, roughly four hundred million of them	Regulates blood flow into a capillary bed and sets resistance

HOW AN ARTERIOLE CONTROLS FLOW

Structure	What it is
Vasoconstriction	The narrowing of a vessel as the smooth muscle of its wall contracts
Vasodilation	The widening of a vessel as the smooth muscle of its wall relaxes
Metarteriole	The tapered terminal end of an arteriole, at the junction with a capillary bed
Precapillary sphincter	A ring of smooth muscle at the metarteriole that opens or closes the entrance to a capillary
Vascular tone	The partial contraction a vessel holds at rest, which stiffens the wall and maintains pressure

Capillaries

Capillaries are the only vessels thin enough for exchange. They lack a tunica media and a tunica externa entirely: the wall is endothelium and a basement membrane. There are three types, leakier as the tissue's job demands.

THE THREE CAPILLARY TYPES

Capillary type	Wall	Where found
Continuous capillary	An unbroken endothelial lining, interrupted only by intercellular clefts	The most common type. Muscle, skin, lungs, and the central nervous system
Fenestrated capillary	An endothelium with small pores, called fenestrations	Where rapid absorption or filtration happens: kidney, small intestine, choroid plexus, most endocrine glands
Sinusoid	A wide, winding vessel with large fenestrations, an incomplete basement membrane, and wide clefts	Where even whole cells must cross: liver, spleen, red bone marrow, anterior pituitary, adrenal glands

HOW CAPILLARIES ARE ORGANISED

Structure	What it is
Capillary bed	A network of ten to one hundred capillaries supplied by a single metarteriole
Microcirculation	The flow from metarteriole to capillaries to postcapillary venule
Thoroughfare channel	The distal end of a metarteriole, which has no smooth muscle and acts as a direct route from arteriole to venule
Vascular shunt	A direct channel that lets blood bypass the capillaries of a bed when exchange is not needed
Avascular tissues	Tissues with no capillaries at all: all covering and lining epithelia, the cornea and lens, and cartilage

HOLD ONTO THIS

Capillary type follows the job of the tissue. Tight where things must be kept out, windowed where fluid must move fast, and frankly leaky where whole cells have to pass. One vessel name, three builds, three jobs.

Veins

Veins return blood to the heart under low pressure. Because the pressure is low, they rely on valves and on nearby muscles to keep blood moving the right way.

THE VENOUS SIDE

Structure	What it is
Venule	The smallest vein, collecting blood as it leaves a capillary bed
Postcapillary venule	The venule that first receives blood from the capillaries. Loosely joined and very porous
Muscular venule	The next vessel along, with one or two layers of smooth muscle
Vein	A vessel returning blood to the heart, with a thinner wall and a wider lumen than the matching artery
Venous valves	Flaps of the tunica intima, common in the limb veins, that keep blood from flowing backward
Skeletal muscle pump	Contracting limb muscles that squeeze the deep veins and push blood toward the heart
Blood reservoirs	The veins as a group, which at rest hold the largest share of the body's blood, roughly sixty percent
Vascular sinus	A vein with a thin endothelial wall and no smooth muscle, supported by surrounding dense connective tissue. The dural venous sinuses and the coronary sinus

HOLD ONTO THIS

A vein has the same three tunics as an artery, in different proportions. Thin media, thick externa, wide lumen. On a slide the artery holds its round shape and the vein looks collapsed, and that is usually the fastest way to tell the pair apart.

Circulatory Routes

Most blood follows the simple loop of artery to capillary to vein, but a few routes are arranged differently.

THE ROUTES WORTH NAMING

Structure	What it is
Pulmonary circuit	The route from the right side of the heart to the lungs and back to the left side
Systemic circuit	The route from the left side of the heart out to the body and back to the right side
Portal system	An arrangement in which blood passes through two capillary beds in series before returning to the heart
Hepatic portal circulation	The portal system that routes blood from the digestive organs through the liver
Hypophyseal portal system	The portal system linking the hypothalamus to the anterior pituitary
Anastomosis	A place where two vessels join, giving blood an alternate route if one path is blocked
Arteriovenous anastomosis	A direct connection from an artery to a vein that bypasses the capillaries

WHY THE WALL BUILT FOR THE JOB MATTERS

Blood is drawn and IV lines are placed in **veins**, not arteries. A vein sits closer to the surface, has a thinner wall and a wider lumen, and runs at low pressure, so it is easy to enter and slow to bleed. Its **valves** can even be seen as small bumps along a forearm vein.

The **capillary** type is matched to the tissue. The leaky **sinusoids** of the liver and spleen let whole blood cells pass, while the tight **continuous capillaries** of the brain keep most substances out. The same vessel name, built three different ways, does three different jobs.

The elastic recoil of the **aorta** keeps blood moving between heartbeats. When elastic arteries stiffen with age, that cushioning is lost, and the strain shows up downstream in the smaller vessels.

CHAPTER 8

Blood Vessel Disorders and Fetal Circulation

BLOCK 3 · MODULE 5: DISORDERS AND THE FETAL PLAN

This chapter covers two topics. First, the common disorders of blood vessels, which mostly narrow or weaken the vessel wall. Second, fetal circulation, the rearranged plan a fetus uses before its lungs work, and what each detour leaves behind after birth.

BY THE END

1. Describe atherosclerosis and how it changes an artery wall.
2. Name common arterial and venous disorders.
3. Explain how fetal circulation differs from the adult plan.
4. Name the fetal shunts and the adult remnant each one becomes.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Blood vessels

drsrennie-stack.github.io/new-build-bio4-solano/vessels-concept-videos.html

An Overview

THE TWO HALVES OF THIS CHAPTER

Structure	What it is
Healthy vessel	A vessel with a smooth endothelial lining and a wall that can stretch and recoil
Vascular disorder	A disease of a blood vessel; most disorders narrow the lumen, weaken the wall, or let blood pool
Fetal circulation	The circulatory plan of a fetus, which gets its oxygen from the placenta rather than from its own lungs
Shunt	A special vessel or opening that routes blood around an organ; the fetus uses shunts to bypass the lungs and liver

Atherosclerosis

Atherosclerosis is the most important vessel disease, the root of heart attacks and strokes. It is a slow, structural change in the artery wall.

THE DISEASE, STRUCTURE BY STRUCTURE

Structure	What it is
Atherosclerosis	A disease in which fatty plaques build up within the walls of arteries
Endothelial injury	Damage to the smooth tunica intima lining, the starting point of a plaque
Atheroma	Also called a plaque, a deposit of fatty material, cells, and debris within the artery wall
Narrowed lumen	As a plaque grows it bulges inward, narrowing the channel and reducing blood flow
Arteriosclerosis	The general stiffening and thickening of artery walls; atherosclerosis is its most common form
Thrombus	A clot that can form on a roughened plaque and suddenly block the vessel
Embolus	A clot or fragment that breaks free and travels through the bloodstream to lodge elsewhere

Arterial Disorders

Arteries carry blood under high pressure, so their disorders tend to narrow the channel or weaken the wall.

COMPARE THE COMMON ONES

Disorder	What it is	Why it matters
Aneurysm	A balloon-like bulge in a weakened artery wall	It can rupture; common in the aorta and the arteries of the brain
Atherosclerosis	Fatty plaque buildup that narrows an artery	The leading cause of coronary artery disease and stroke
Coronary artery disease	Narrowing of the coronary arteries that supply the heart wall	Can starve the myocardium and cause a heart attack
Peripheral artery disease	Narrowing of the arteries of the limbs, usually the legs	Causes pain on walking as the muscles are starved of blood
Hypertension	Chronically high pressure within the arteries	Strains the heart and damages vessel walls over time
Stroke	Loss of blood supply to part of the brain, from a blocked or burst artery	Brain tissue begins to die within minutes

Venous Disorders

Veins carry blood under low pressure, so their disorders tend to involve failed valves and pooled or clotted blood.

COMPARE THE COMMON ONES

Disorder	What it is	Why it matters
Varicose veins	Superficial veins that become twisted and swollen when their valves fail	Blood pools and the veins bulge visibly, most often in the legs
Deep vein thrombosis	A blood clot in a deep vein, usually of the leg	The clot can break free and travel through the bloodstream
Pulmonary embolism	A clot, often from a deep vein, that lodges in an artery of the lung	A life-threatening emergency that blocks blood flow to the lung
Phlebitis	Inflammation of a vein	Causes pain, redness, and tenderness along the vein
Hemorrhoids	Varicose veins of the rectal and anal region	A common cause of rectal discomfort and bleeding
Chronic venous insufficiency	Long-term failure of the valves of the leg veins	Leads to persistent swelling and skin changes in the lower leg

Fetal Circulation, an Overview

Before birth, a fetus does not breathe air or digest food, so its circulation is rearranged. Oxygen and nutrients come from the placenta, and blood is routed around the organs that are not yet in use.

WHAT IS DIFFERENT BEFORE BIRTH

Structure	What it is
The placenta	The organ where fetal blood picks up oxygen and nutrients from the mother and gives up wastes
Umbilical cord	The cord connecting the fetus to the placenta, carrying the umbilical vessels
The lungs are bypassed	The fetal lungs are collapsed and not used for gas exchange, so most blood is routed around them
The liver is largely bypassed	Much fetal blood skips the liver, since the placenta handles that filtering
Three shunts	The ductus venosus, the foramen ovale, and the ductus arteriosus, the detours that route blood around the liver and lungs

The Fetal Vessels and Shunts

Five special structures carry and reroute fetal blood. Compare what each one is and what it does.

THE FETAL STRUCTURES

Structure	What it is	What it does
Umbilical vein	A vein in the umbilical cord	Carries oxygen-rich blood from the placenta to the fetus
Ductus venosus	A shunt passing through the fetal liver	Lets most blood bypass the liver and flow straight toward the inferior vena cava
Foramen ovale	An opening in the interatrial septum	Lets blood pass directly from the right atrium to the left atrium, bypassing the lungs
Ductus arteriosus	A short vessel between the pulmonary trunk and the aorta	Lets blood bypass the lungs by shunting it from the pulmonary trunk into the aorta
Umbilical arteries	Two arteries in the umbilical cord	Carry oxygen-poor blood from the fetus back to the placenta

The Shunts After Birth

At birth the lungs inflate and the placenta is cut off. Each fetal shunt is no longer needed, closes, and is left behind as a small ligament or remnant.

COMPARE EACH ONE WITH WHAT IT BECOMES

Fetal structure	Adult remnant	Note
Umbilical vein	The ligamentum teres, the round ligament of the liver	Runs in the free edge of the falciform ligament
Ductus venosus	The ligamentum venosum	A fibrous cord on the inferior surface of the liver
Foramen ovale	The fossa ovalis	A shallow oval depression in the interatrial septum
Ductus arteriosus	The ligamentum arteriosum	A short cord between the pulmonary trunk and the aorta
Umbilical arteries	The medial umbilical ligaments	Paired cords on the inner surface of the anterior abdominal wall

HOLD ONTO THIS

Every fetal shunt becomes a ligament, and the name usually keeps a piece of the original. Ductus venosus becomes ligamentum venosum, ductus arteriosus becomes ligamentum arteriosum. The one that breaks the pattern is the foramen ovale, which becomes a depression rather than a cord, because it was a hole and not a vessel.

WHEN THE DETOUR DOES NOT CLOSE

The fetal shunts are meant to close in the first hours and weeks after birth. When the **ductus arteriosus** stays open, the result is a patent ductus arteriosus; when the **foramen ovale** stays open, a patent foramen ovale. Both leave an abnormal connection between the two sides of the circulation, and both are named for the fetal structure that failed to seal.

Atherosclerosis ties the disorders together. A plaque that narrows a coronary artery causes a heart attack; one that narrows or bursts a brain artery causes a stroke; one in the leg causes peripheral artery disease. The same wall change, in different vessels, names different diseases.

A **deep vein thrombosis** in the leg is dangerous for where the clot can go. If it breaks free, it travels up the venae cavae, through the right heart, and lodges in a lung artery as a **pulmonary embolism**. The route the clot takes is just the normal venous return.

CHAPTER 9

Blood

BLOCK 3 · MODULE 1: COMPOSITION, FORMATION, AND DISORDERS

A reference for the blood video and lab. This chapter covers the composition of blood, plasma and the formed elements, how blood cells are formed, and common blood disorders. The focus is on structure and the named components.

BY THE END

1. Describe the composition of blood: plasma and the formed elements.
2. Identify the structure of erythrocytes, the five leukocytes, and platelets.
3. Explain where and how blood cells are formed.
4. Name common blood disorders and the component each one affects.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Blood

drsrennie-stack.github.io/new-build-bio4-solano/blood-concept-videos.html

Blood, an Overview

Blood is a connective tissue, the body's only liquid tissue. Like every connective tissue it is cells suspended in a matrix; here the matrix, plasma, is a fluid.

COMPOSITION AND PHYSICAL PROPERTIES

Structure	What it is
Blood	A fluid connective tissue that transports materials throughout the body
Plasma	The liquid extracellular matrix of blood, roughly fifty five percent of blood volume
Formed elements	The cells and cell fragments suspended in the plasma, roughly forty five percent of blood volume
Volume	Seven to eight percent of body weight, about five litres in an average adult
Temperature	About thirty eight degrees Celsius, slightly warmer than core body temperature
Viscosity	About five times more viscous than water, from the protein content and the formed elements
Hematocrit	The percentage of blood volume made up of erythrocytes
Buffy coat	The thin pale layer of leukocytes and platelets between the plasma and the erythrocytes in a spun sample
Functions of blood	Transport of gases, nutrients, wastes, and hormones; regulation of temperature and pH; protection through clotting and immune defense

Plasma

Plasma is about ninety two percent water. The rest is dissolved solutes, and the plasma proteins are the most abundant of them. All but the gamma globulins are made in the liver.

THE THREE MAIN PLASMA PROTEINS

Plasma protein	Share	Made by	Role
Albumin	About sixty percent of plasma protein	The liver	The most abundant plasma protein. Maintains the osmotic pressure that holds water in the blood, and carries hormones, fatty acids, and drugs
Globulins	About thirty five percent	The liver and plasma cells	Alpha and beta globulins are transport proteins. Gamma globulins are the antibodies of immune defense
Fibrinogen	About four percent	The liver	The largest plasma protein and the clotting protein. It converts to fibrin threads that form the framework of a clot

THE REST OF PLASMA

Structure	What it is
Water	About ninety two percent of plasma, the solvent that carries every other component
Nutrients	Glucose, amino acids, fatty acids, and vitamins
Wastes	Urea, creatinine, bilirubin, and ammonia
Electrolytes	Sodium, potassium, calcium, chloride, and bicarbonate
Gases	Dissolved oxygen, carbon dioxide, and nitrogen
Serum	Plasma with the clotting proteins removed, the fluid left after blood has clotted

The Formed Elements

The formed elements are the cellular part of blood. There are three kinds, and only one of them is a complete cell type with a nucleus.

COMPARE THEM

Formed element	Also called	Description	Relative number
Erythrocytes	Red blood cells	Biconcave discs without a nucleus, packed with hemoglobin	By far the most numerous
Leukocytes	White blood cells	Complete nucleated cells that defend the body	The least numerous
Platelets	Thrombocytes	Small cell fragments, not whole cells, that act in clotting	Intermediate in number

Erythrocytes

Erythrocytes, the red blood cells, carry oxygen. Almost every feature of their structure is a compromise made to fit more hemoglobin into the cell.

STRUCTURE AND ITS CONSEQUENCES

Structure	What it is
Erythrocyte	A red blood cell, the oxygen carrier of the blood
Biconcave disc	The erythrocyte's shape, a disc thinner at its center. About seven and a half micrometres across, giving thirty percent more surface area than a sphere of the same volume
No nucleus	A mature erythrocyte has ejected its nucleus and most organelles, leaving more room for hemoglobin
Flexible plasma membrane	Lets the disc bend and fold to squeeze single file through the narrowest capillaries
Hemoglobin	The red, iron-containing protein that fills the erythrocyte and binds oxygen for transport. Four subunits, each with a heme group holding one iron
Lifespan	About one hundred and twenty days, after which worn cells are broken down by the spleen and liver

HOLD ONTO THIS

Every structural oddity of the red cell serves one job. No nucleus means more room for hemoglobin. A biconcave shape means more surface for gas exchange. A flexible membrane means it can fold through a capillary narrower than itself. Structure follows function here more plainly than anywhere else in the course.

Leukocytes

Leukocytes, the white blood cells, defend the body. They sort into granulocytes, which have visible cytoplasmic granules, and agranulocytes, which do not.

THE FIVE TYPES

Leukocyte	Group	Appearance	Defense role
Neutrophil	Granulocyte	A multi-lobed nucleus, three to five lobes, and fine pale granules. The most numerous leukocyte	The first responder to infection. Engulfs and digests bacteria
Eosinophil	Granulocyte	A two-lobed nucleus and coarse red-orange granules	Attacks parasites and moderates allergic reactions
Basophil	Granulocyte	A lobed nucleus hidden by large dark granules. The rarest leukocyte	Releases histamine and other mediators in inflammation
Lymphocyte	Agranulocyte	A large round nucleus filling almost the whole cell	The core of specific immunity. Includes the T cells and B cells
Monocyte	Agranulocyte	The largest leukocyte, with a kidney-shaped nucleus	Leaves the blood and matures into a macrophage that engulfs debris and pathogens

HOLD ONTO THIS

Never Let Monkeys Eat Bananas: Neutrophils, Lymphocytes, Monocytes, Eosinophils, Basophils, in order from most to least numerous. The order is the answer to the differential count question.

Platelets and Blood Formation

Platelets are not whole cells, and all the formed elements trace back to one stem cell in the red bone marrow. The formation of blood cells is called hematopoiesis.

PLATELETS

Structure	What it is
Platelet	Also called a thrombocyte, a small membrane-bound fragment of cytoplasm, not a complete cell
Megakaryocyte	The giant bone marrow cell whose cytoplasm breaks into thousands of platelets
Role in clotting	Platelets gather and stick at a vessel injury, forming a plug and helping the clotting proteins build a clot

HEMATOPOIESIS

Structure	What it is
Hematopoiesis	The formation of the blood's formed elements
Red bone marrow	The site of hematopoiesis in adults, found in the spongy bone of the axial skeleton and the proximal limb bones
Hemocytoblast	The pluripotent hematopoietic stem cell that every formed element arises from
Myeloid line	The line that gives rise to erythrocytes, platelets, and all the leukocytes except lymphocytes
Lymphoid line	The line that gives rise to the lymphocytes
Reticulocyte	An immature erythrocyte with ribosome remnants still in it. Normally one to two percent of circulating red cells
Erythropoietin	The hormone, made chiefly by the kidney, that stimulates erythrocyte production when tissues are short of oxygen

Blood Disorders

Most blood disorders are a problem of one formed element or one plasma protein, too few, too many, or abnormal.

COMPARE THEM BY THE COMPONENT EACH ONE AFFECTS

Disorder	Component affected	What it is
Anemia	Erythrocytes	Too few erythrocytes or too little hemoglobin, so the blood carries less oxygen
Sickle cell disease	Erythrocytes and hemoglobin	An inherited abnormal hemoglobin distorts erythrocytes into a sickle shape that jams capillaries
Polycythemia	Erythrocytes	An excess of erythrocytes that thickens the blood and slows its flow
Leukemia	Leukocytes	A cancer of white blood cell formation that crowds the marrow with abnormal leukocytes
Leukopenia	Leukocytes	An abnormally low white blood cell count, which weakens the body's defense
Thrombocytopenia	Platelets	An abnormally low platelet count, which raises the risk of bleeding
Hemophilia	Clotting proteins	An inherited deficiency of a clotting factor, so blood clots too slowly

ONE TUBE OF BLOOD, A WHOLE STATUS REPORT

Because blood is a tissue that can be sampled with a needle, a single tube reports on the whole body. Spinning it gives the **hematocrit**, the fraction of red cells; a complete blood count tallies each formed element. A low red cell count points to **anemia**, a high white cell count toward infection or **leukemia**.

The structure of the erythrocyte explains its diseases. In **sickle cell disease** a single abnormal hemoglobin makes the cell rigid and curved, so it cannot fold through a capillary and blocks it instead, causing pain and tissue damage downstream.

Because **hematopoiesis** happens in red bone marrow, a marrow sample drawn from the iliac crest or the sternum is how the blood-forming tissue itself is examined when a count is abnormal.

PART B

Pre–work.

Four packets, in the order they are due. Work them in sequence: read the Part A chapter, organize it into mini visuals, draw it from memory, apply it with your notes closed, then retrieve later in the week by redrawing from memory. Context before content. This is not a night-before task.

1. **Muscle and the Muscles of the Trunk and Upper Limb**
2. **The Heart**
3. **Blood Vessels and Blood**
4. **Vessels and Nerves of the Upper Limb**

THESE PAGES PRINT LANDSCAPE. TURN THE PACKET SIDEWAYS.

Pre-work · Muscle and the Muscles of the Trunk and Upper Limb

Muscle Structure and Sarcomeres, and Fascicle Arrangement and Lever Systems. Read both Part A chapters first. The last two visuals and the last drawing are lab prep: the named muscles are anatomy lab material, so pull their origin, insertion, and action from the muscle chart on the Loops page, not from Part A.

Module 3 · 1 of 4

Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the two muscle chapters into mini visuals, then use the last two to get ahead of lab. Eight chunks. The grids carry the naming, the hubs carry the structure.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 **LADDER**

Levels of muscle organization

Five rungs, muscle at the top down to myofilament at the bottom. Write the wrapping beside each rung that has one. Three rungs get a sheath and two do not, and that is the point.

2 **HUB**

The sarcomere

One sarcomere across the middle, Z disc to Z disc. Spoke out to A band, I band, H zone, M line, and zone of overlap. Shade which filaments sit in each. This is the most tested figure in the chapter.

3 **SPLIT**

Thick against thin filament

Main protein, structure, and what it anchors to. Hang tropomyosin and troponin off the thin branch, titin and myomesin off the thick.

4 **GRID**

The three muscle tissues

Skeletal, cardiac, smooth down the side. Location, striations, control, and cell features across the top. Star the two features that are unique to one tissue each.

5 **GRID**

Fascicle arrangement

Pattern down the side, arrangement, example muscle, and power against range across the top. Six rows. The last column is the one that answers application questions.

DRAWING 1

A sarcomere, fully labelled

BRAIN DUMP FIRST

Two minutes. Every band, line, and zone, and which filament sits where.

Draw one sarcomere large, Z disc to Z disc. Lay in the thick filaments anchored at the M line and the thin filaments anchored at both Z discs. Now shade and label the A band, both I bands, the H zone, the M line, and the zone of overlap. Add titin running Z disc to M line.

In your notebook, head the page **M3-1 A sarcomere, fully labelled** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say which regions narrow when the muscle shortens, and which one never changes, and point to why on your own drawing.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A muscle has short fascicles set at an angle on both sides of a central tendon. Predict its power and its range of motion, and name a muscle built this way.

Answer. High power, small range of motion. Bipennate. Rectus femoris is an example.

Reasoning. Start with the geometry, not the name. Angling the fascicles lets far more of them pack into the same volume of muscle, and every one of them pulls on the tendon, so force adds up. But each fascicle is short, and a fiber can only shorten to about two thirds of its resting length, so the tendon does not travel far. Power and range trade against each other, and the architecture is where the trade is made visible.

Set A · Name it

1. The sheath wrapping a single muscle fiber is the ____________________.
2. The structural unit of one T tubule flanked by two terminal cisternae is a ____________________.
3. The band that contains thin filaments only is the ____________________.
4. The elastic protein running from the Z disc to the M line is ____________________.
5. Name the three lever classes and one body example of each.

____

____

Set B · Predict it

1. A muscle is rebuilt from parallel to pennate architecture. Predict what happens to its power and to its range.

____

____

6 **LADDER**

The three lever classes

Three rungs, first through third. Draw the bar with fulcrum, effort, and load in position on each rung, then write the body example beside it.

7 **GRID**

Superficial back and anterior thorax, lab prep

Muscle down the side, origin, insertion, action, and innervation across the top. Nine rows, filled from the muscle chart on the Loops page. Then add one more column: which scapular movement each produces, and circle the antagonist pairs.

8 **HUB**

The rotator cuff and the shoulder, lab prep

Head of the humerus at the hub. Spoke to supraspinatus, infraspinatus, teres minor, subscapularis, and mark the tubercle each reaches. Add deltoid, teres major, and coracobrachialis around the outside. Origin, insertion, and action come from the muscle chart.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

A muscle in cross section, from sheath to fiber

BRAIN DUMP FIRST

One minute. The three sheaths, what each wraps, and how they reach the tendon.

Draw a muscle cut across. Outline the whole muscle with the epimysium, divide the inside into fascicles each ringed by perimysium, and draw a handful of fibers inside one fascicle each ringed by endomysium. Then draw the muscle from the side and show all three sheaths merging into the tendon at the end.

In your notebook, head the page **M3–2 A muscle in cross section, from sheath to fiber** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can trace a line with your finger from inside one fiber all the way to the bone without lifting it off the page.

2. A tendon is torn from the bone. Predict how the pull of the individual fibers is affected, using the three sheaths.

3. Dystrophin is faulty in a muscle fiber. Predict what happens with ordinary use, and say why.

4. A tendon is moved surgically so it inserts farther from the joint. Predict the effect on strength and on speed.

Set C · Tell it apart

1. Distinguish a muscle strain from a tendon rupture by which structure fails and how each heals.

2. Compare the three muscle tissues by striations and control, and name the one feature that places each on a slide.

3. Explain how the banding of a sarcomere produces the striations you see down a microscope.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The posterior shoulder and back, in layers, lab prep

BRAIN DUMP FIRST

Two minutes. Every superficial back muscle and the rotator cuff underneath.

Draw the back of the trunk and scapula. Lay in trapezius and latissimus dorsi as the superficial layer on the left half. On the right half draw them peeled away to show levator scapulae, both rhomboids, and the rotator cuff on the scapula. Mark the origin and insertion of each with a dot and a line.

In your notebook, head the page **M3-3 The posterior shoulder and back, in layers, lab prep** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name what happens to the scapula when each muscle you drew contracts, without reading the label.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Levels of muscle organization

LADDER

2 The sarcomere

HUB

3 Thick against thin filament

SPLIT

4 The three muscle tissues

GRID

5 Fascicle arrangement

GRID

6 The three lever classes

LADDER

7 Superficial back and anterior thorax, lab prep

GRID

8 The rotator cuff and the shoulder, lab prep

HUB

Pre-work · The Heart

The Heart and the Cardiac Conduction System. Read both Part A chapters first. Almost every question on this material is a pathway question, so the order is the content.

Module 3 · 2 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the heart and conduction chapters into mini visuals. Six chunks. Two are chains, and those two carry most of the exam weight.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 LADDER

The heart wall and pericardium

Five rungs, from fibrous pericardium on the outside down to endocardium on the inside. Write beside the epicardium rung the second name it goes by, because that is the one people miss.

2 GRID

The four chambers

Chamber down the side. Receives from, pumps to, wall thickness, and one internal feature across the top. Four rows.

3 GRID

The four valves

Valve down the side. Type, number of cusps, location, and what it prevents across the top. Add a fifth column: has chordae, yes or no.

4 CHAIN

The pathway of blood, nine steps

Nine boxes with arrows, right atrium through to the aorta. Colour the oxygen-poor half one way and the oxygen-rich half the other, and mark exactly where the switch happens.

5 CHAIN

The conduction pathway, six steps

SA node through to Purkinje fibers, arrows the whole way. Mark where the delay happens and write beside it what the delay buys.

DRAWING 1

The heart, frontal section, with flow arrows

BRAIN DUMP FIRST

Two minutes. Every chamber, valve, and great vessel, and the direction blood moves.

Draw the heart cut front to back so all four chambers show. Add the interatrial and interventricular septa, all four valves in position, chordae tendineae and papillary muscles in both ventricles, and the five great vessels attached where they belong. Now draw arrows through the whole path and shade oxygen-poor and oxygen-rich blood differently.

In your notebook, head the page **M3-1 The heart, frontal section, with flow arrows** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can start at any point in your drawing and say what comes next, forward or backward, without looking at the notes.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A patient has a murmur heard when the ventricles contract, and imaging shows a torn papillary muscle in the left ventricle. Name the valve involved and explain the sound.

Answer. The mitral valve.

Reasoning. Work from the structure named. Papillary muscles exist only in the ventricles and anchor only the atrioventricular valves, so a semilunar valve is ruled out immediately. The left ventricle narrows it to the mitral valve. When the ventricle contracts, the pressure behind that valve is enormous, and the chordae tendineae held by that papillary muscle are the only thing stopping the cusps from being pushed back into the atrium. Torn anchor, cusps flip, blood slips backward, and the turbulence is the murmur. Notice the sound is timed to ventricular contraction, which is itself a clue.

Set A · Name it

1. The visceral layer of the serous pericardium is also called the ________________________.
2. The vessel returning oxygen-rich blood to the left atrium is the ________________________.
3. The remnant of the fetal foramen ovale in the adult atrium is the ________________________.
4. The only electrical bridge between the atria and the ventricles is the ________________________.
5. Name the two coronary arteries and one main branch of each.

   ____

Set B · Predict it

1. The anterior interventricular artery is blocked. Predict which region of the heart is damaged and why.

   ____

   ____

6 **HUB**

Coronary circulation

Aorta at the hub. Two arteries off it, then their named branches, then the matching cardiac veins running back to the coronary sinus. Draw the vein alongside the artery it travels with.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The conduction system on a heart outline

BRAIN DUMP FIRST

One minute. Every component in order, and where each one sits.

Draw a plain heart outline. Mark the SA node in the right atrial wall near the superior vena cava, the AV node in the atrial floor, the AV bundle in the interventricular septum, both bundle branches, and the Purkinje fibers spreading through the ventricle walls. Draw the fibrous skeleton as a band between atria and ventricles and mark the one place the signal can cross.

In your notebook, head the page **M3–2 The conduction system on a heart outline** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain from your drawing why a block at the AV bundle disconnects the atria from the ventricles entirely.

2. Fluid collects in the pericardial cavity. Predict the effect on filling, and name the condition.

3. The delay at the AV node is lost. Predict what happens to ventricular filling.

4. A right coronary artery occlusion causes a rhythm problem as well as muscle damage. Explain why, anatomically.

Set C · Tell it apart

1. Distinguish the atrioventricular from the semilunar valves by structure and by what holds each shut.

2. Explain why the left ventricle wall is thicker than the right, using the circuit each one drives.

3. A transplanted heart still beats though its nerves were cut. Explain what that proves.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

An ECG tracing lined up with the pathway

BRAIN DUMP FIRST

One minute. The three deflections and the conduction event behind each.

Draw one heartbeat of an ECG: P wave, QRS complex, T wave. Under each deflection write the conduction event that produces it and the chamber action that follows. Then draw a second tracing in a different colour with the P waves marching independently of the QRS complexes, and label what that is.

In your notebook, head the page **M3-3 An ECG tracing lined up with the pathway** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can point at any part of the tracing and name what the conduction system is doing at that moment.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The heart wall and pericardium LADDER

2 The four chambers GRID

3 The four valves GRID

4 The pathway of blood, nine steps

CHAIN

5 The conduction pathway, six steps

CHAIN

6 Coronary circulation

HUB

Pre-work · Blood Vessels and Blood

Blood Vessels, Blood Vessel Disorders and Fetal Circulation, and Blood. Read all three Part A chapters first. Most of this is histology, so work it with a slide or an image beside you.

Module 3 · 3 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the vessel and blood chapters into mini visuals. Eight chunks. Two grids do the histology and one chain does the fetal detour.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 GRID

The three tunics

Tunic down the side. Position, tissue, and one distinguishing note across the top. Add a fourth column: is it present in a capillary, yes or no.

2 GRID

The five vessel types

Vessel down the side, in the order blood meets them. Direction, wall, and role across the top. This grid is the frame for the whole chapter.

3 SPLIT

Artery against vein, on a slide

One split. Wall thickness, lumen shape, tunica media, tunica externa, and valves. This is the pair you will be asked to tell apart, so make the giveaway column obvious.

4 GRID

The three capillary types

Capillary down the side, wall and where found across the top. Add a fourth column: what can cross it. Three rows, and the third column is the reasoning column.

5 CHAIN

Fetal circulation, placenta to placenta

One long chain with arrows. Umbilical vein, ductus venosus, right atrium, then the two ways past the lungs, and back out the umbilical arteries. Mark each shunt where it appears.

DRAWING 1

An artery and a vein, side by side in cross section

BRAIN DUMP FIRST

One minute. All three tunics in each, and every difference between them.

Draw the two vessels beside each other, cut across. Label all three tunics in each, and draw the internal and external elastic laminae where they belong. Make the artery wall thick with a round lumen and the vein wall thin with a wide, collapsed lumen. Add a valve to the vein and vasa vasorum to the externa of both.

In your notebook, head the page **M3–1 An artery and a vein, side by side in cross section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

Someone shows you an unlabelled slide and you name the vessel in under five seconds, and can say which feature decided it.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A slide shows a vessel with a wide, irregular lumen, an incomplete basement membrane, and very large gaps between the endothelial cells. Whole blood cells are passing through the wall. Name the vessel and one organ it is found in.

Answer. A sinusoid. Liver, spleen, or red bone marrow.

Reasoning. Only one vessel in the body has an incomplete basement membrane, and that alone settles it. Work back from what is crossing: a continuous capillary keeps most substances out, a fenestrated capillary passes fluid and small solutes through pores, but whole cells need gaps far larger than either. So the tissue must be one where cells legitimately move between blood and tissue, which is exactly what the liver, spleen, and marrow do. Structure is matched to the job, and the leakiest wall goes where the largest thing has to cross.

Set A · Name it

1. The only tunic a capillary has is the ____________________.
2. The small vessels supplying the wall of a large vessel are the ____________________.
3. The ring of smooth muscle controlling entry to a capillary is the ____________________.
4. The adult remnant of the ductus arteriosus is the ____________________.
5. Name the three main plasma proteins and one job of each.

____

____

Set B · Predict it

1. A deep vein thrombosis forms in the calf and breaks free. Trace the exact route it takes and name where it lodges.

6 GRID

Fetal shunt to adult remnant

Fetal structure down the side, what it does and what it becomes across the top. Five rows. Circle the one that does not become a cord.

7 HUB

Blood, composition

One spun tube at the hub. Spoke to plasma with its three proteins, the buffy coat, and the erythrocyte layer. Write the percentage beside each layer.

8 GRID

The five leukocytes

Leukocyte down the side. Group, nucleus, granules, and defense role across the top. Order the rows most to least numerous, because that order is itself an answer.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The fetal circulation, with the three shunts

BRAIN DUMP FIRST

Two minutes. Every fetal vessel and where each detour goes.

Draw a simple fetus with the heart, liver, and lungs marked, and the placenta beside it. Draw the umbilical vein in, the ductus venosus past the liver, the foramen ovale between the atria, the ductus arteriosus between the pulmonary trunk and the aorta, and the two umbilical arteries out. Arrow the whole route. Then in a second colour write the adult remnant beside each shunt.

In your notebook, head the page **M3–2 The fetal circulation, with the three shunts** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain why each shunt exists by pointing at the organ it bypasses and saying what that organ is not yet doing.

2. The valves of the leg veins fail. Predict what you would see and explain it from venous pressure.
3. The ductus arteriosus fails to close after birth. Predict the abnormal connection this leaves.
4. An abnormal hemoglobin makes a red cell rigid. Predict where in the circulation it causes trouble and why.

Set C · Tell it apart

1. Explain why blood is drawn from a vein rather than an artery, using three structural differences.
2. Distinguish an aneurysm from an atherosclerotic plaque by what each does to the vessel wall.
3. Blood is called a connective tissue. Defend that classification using its matrix.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

A blood smear

BRAIN DUMP FIRST

One minute. Every formed element, drawn as it looks under the microscope.

Draw a field of a blood smear. Fill most of it with biconcave erythrocytes, showing the pale centre. Add one of each leukocyte, drawn by its nucleus: multi-lobed neutrophil, bilobed eosinophil, granule-obscured basophil, round-nucleus lymphocyte, kidney-shaped monocyte. Scatter platelets between them. Label each and note its relative number.

In your notebook, head the page **M3-3 A blood smear** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can identify each cell from the nucleus alone, with the labels covered.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The three tunics
GRID

2 The five vessel types
GRID

3 Artery against vein, on a slide
SPLIT

4 The three capillary types
GRID

5 Fetal circulation, placenta to placenta
CHAIN

6 Fetal shunt to adult remnant

GRID

7 Blood, composition

HUB

8 The five leukocytes

GRID

Pre-work · Vessels and Nerves of the Upper Limb

Vessels and Nerves of the Thorax and Upper Limb. This is anatomy lab material, so there is no Part A chapter for it. Work from the Part C structure list and from lab. These are routes, not lists, so trace them with your finger before you try to name anything.

Module 3 · 4 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the Part C vessel and nerve lists into mini visuals. Six chunks. Chains for the routes, grids for the nerves.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 **CHAIN**

Aortic arch to the hand

One chain with arrows, from the arch through subclavian, axillary, brachial, then splitting to radial and ulnar and joining as the palmar arches. Write the landmark at every name change, because the landmark is the answer.

2 **SPLIT**

Right side against left side, off the arch

Two branches. Show why the right arm's supply needs an extra vessel and the left arm's does not. Name all three branches of the arch on the correct side.

3 **GRID**

Superficial against deep veins

Vein down the side. System, course, and where it joins across the top. Eight rows. Star the one used for a blood draw.

4 **LADDER**

The brachial plexus, five levels

Five rungs, roots to branches. Write the count at each rung: five roots, three trunks, six divisions, three cords, five terminal branches. The counts are what tell you if you have missed one.

5 **GRID**

The five terminal nerves

Nerve down the side. Cord of origin, what it supplies, and the deficit its injury causes across the top. This grid answers most clinical questions on the chapter.

DRAWING 1

The arterial tree of the upper limb

BRAIN DUMP FIRST

Two minutes. Every named artery from the arch to the hand, in order.

Draw the trunk outline and one arm. Start at the aortic arch and draw all three branches, correctly asymmetrical. Follow the right side out: subclavian, axillary, brachial, then radial and ulnar, then both palmar arches. Mark the first rib, the lower border of teres major, and the cubital fossa as the three landmarks where names change. Add the deep brachial artery in the radial groove.

In your notebook, head the page **M3–1 The arterial tree of the upper limb** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name any artery on your drawing and say what it was called one segment earlier and what landmark changed it.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A patient fractures the shaft of the humerus and afterwards cannot extend the wrist or the fingers, though elbow flexion is intact. Name the nerve and explain the pattern.

Answer. The radial nerve.

Reasoning. Start with what is lost, not with the fracture. Wrist and finger extension is the job of the posterior compartment of the forearm, and one nerve supplies that whole compartment. Elbow flexion is intact, so the musculocutaneous nerve in the anterior compartment of the arm is fine, which rules it out. Now the fracture site confirms it: the radial nerve spirals around the shaft of the humerus in the radial groove, pressed directly against the bone, so a mid-shaft break and this nerve are in the same place. Compartment gives you the nerve, and the bone tells you where it was injured.

Set A · Name it

1. The subclavian artery becomes the axillary artery at the ____________________.
2. The vein most often used for drawing blood at the elbow is the ____________________.
3. The nerve supplying serratus anterior is the ____________________.
4. The cord that gives rise to the radial and axillary nerves is the ____________________.
5. Name the three branches of the aortic arch, in order.

Set B · Predict it

1. A shoulder dislocation damages a nerve at the surgical neck of the humerus. Name it and predict the deficit.

6 HUB

The humerus and the nerves that hug it

One humerus in the middle. Spoke to axillary at the surgical neck, radial in the radial groove, ulnar behind the medial epicondyle, median in the cubital fossa. Write the injury beside each spoke.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The brachial plexus, roots to branches

BRAIN DUMP FIRST

Two minutes. All five levels and the five terminal nerves.

Draw C5 through T1 as five lines on the left. Merge them into three trunks, split each trunk into two divisions, gather the divisions into three cords, and bring the cords to the five terminal nerves. Label every level. Then draw the axillary artery through the middle and show why the cords are named lateral, posterior, and medial.

In your notebook, head the page **M3–2 The brachial plexus, roots to branches** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can cover your labels and rebuild the whole plexus from the counts alone: five, three, six, three, five.

2. A patient knocks the medial epicondyle. Predict which fingers tingle and explain why.
3. The radial artery is used for an arterial line. Explain, anatomically, why the hand stays supplied.
4. A blood draw at the cubital fossa is taken too deep. Name the two structures immediately beneath the vein.

Set C · Tell it apart

1. Explain why there is one brachiocephalic artery but two brachiocephalic veins.
2. Distinguish the cephalic from the basilic vein by side and by where each ends.
3. Compare a median nerve injury at the wrist with an ulnar nerve injury at the elbow, by muscles lost and by hand appearance.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

A cross section of the arm

BRAIN DUMP FIRST

One minute. Both compartments, with their vessels and nerves in place.

Draw the arm cut across at mid-shaft. Put the humerus in the middle and divide the surrounding tissue into anterior and posterior compartments with the intermuscular septa. Fill in the muscles of each. Then place the brachial artery and median nerve in the anterior compartment and the radial nerve with the deep brachial artery in the radial groove against the bone.

In your notebook, head the page **M3–3 A cross section of the arm** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain from your section why a mid-shaft humeral fracture produces wrist drop and not weak elbow flexion.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Aortic arch to the hand CHAIN

2 Right side against left side, off the arch SPLIT

3 Superficial against deep veins GRID

4 The brachial plexus, five levels

LADDER

5 The five terminal nerves

GRID

6 The humerus and the nerves that hug it

HUB

PART C

Lab Structure List.

Part C of the Module 3 packet. These are the lab structures and concepts to learn for the lab portion of the Module 3 exam. Treat it as the master list, and cross reference it against every lab worksheet. Be prepared to identify structures on models, images, the cadaver, and slides.

CHAPTER 1

Muscle Tissue and Microanatomy

SLIDES AND THE SARCOMERE MODEL

Three tissues on the slide, then the sarcomere on the model. Fill in function and location for each tissue as you go; those columns are what the practical asks you to reason from.

☐ Skeletal muscle, longitudinal section

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Muscle fiber
- ☐ Sarcolemma
- ☐ Sarcoplasm
- ☐ Peripheral nuclei
- ☐ Striations
- ☐ Myofibrils
- ☐ Endomysium

☐ Skeletal muscle, cross section

STRUCTURES TO IDENTIFY

- ☐ Muscle fiber, cut across
- ☐ Fascicle
- ☐ Endomysium
- ☐ Perimysium
- ☐ Epimysium
- ☐ Peripheral nuclei

☐ Cardiac muscle

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Cardiomyocyte
- ☐ Branching fibers
- ☐ Central nucleus
- ☐ Intercalated discs
- ☐ Striations

☐ Smooth muscle

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Spindle-shaped cell
- ☐ Central nucleus
- ☐ Absence of striations
- ☐ Longitudinal layer
- ☐ Circular layer

☐ The muscle organ model

STRUCTURES TO IDENTIFY

- ☐ Epimysium
- ☐ Perimysium
- ☐ Endomysium
- ☐ Fascicle
- ☐ Muscle fiber
- ☐ Tendon
- ☐ Aponeurosis
- ☐ Deep fascia
- ☐ Belly
- ☐ Origin
- ☐ Insertion

☐ The muscle fiber model

STRUCTURES TO IDENTIFY

- ☐ Sarcolemma
- ☐ Sarcoplasm
- ☐ Myonuclei
- ☐ Myofibril
- ☐ Transverse (T) tubule
- ☐ Sarcoplasmic reticulum
- ☐ Terminal cisternae
- ☐ Triad
- ☐ Mitochondria

☐ The sarcomere model

Be prepared to state which filaments are present in each band, line, and zone.

STRUCTURES TO IDENTIFY

- ☐ Z disc
- ☐ A band
- ☐ I band
- ☐ H zone
- ☐ M line
- ☐ Zone of overlap
- ☐ Thick filament (myosin)
- ☐ Thin filament (actin)
- ☐ Titin
- ☐ Cross bridges

Which regions of the sarcomere narrow when a muscle shortens? ______________________________

Which region never changes length, and why? ______________________________

☐ Fascicle arrangement, on models

Name one example muscle for each pattern.

STRUCTURES TO IDENTIFY

- ☐ Parallel
- ☐ Fusiform
- ☐ Circular
- ☐ Convergent
- ☐ Unipennate
- ☐ Bipennate
- ☐ Multipennate

☐ Lever systems

Be prepared to classify a body movement by lever class and to name the fulcrum, effort, and load in it.

STRUCTURES TO IDENTIFY

- ☐ Fulcrum
- ☐ Effort
- ☐ Load
- ☐ First class lever
- ☐ Second class lever
- ☐ Third class lever

CHAPTER 2

Muscles of the Back and Thorax

TRUNK MUSCULATURE

Work these in layers, superficial to deep. For every muscle be prepared to give the origin, the insertion, and the action, not just the name.

☐ Superficial back

STRUCTURES TO IDENTIFY

- ☐ Trapezius
- ☐ Latissimus dorsi
- ☐ Levator scapulae
- ☐ Rhomboid major
- ☐ Rhomboid minor
- ☐ Thoracolumbar fascia
- ☐ Ligamentum nuchae

☐ Erector spinae

Identify them in order from lateral to medial. Be prepared to name the column from its position alone.

STRUCTURES TO IDENTIFY

- ☐ Iliocostalis
- ☐ Longissimus
- ☐ Spinalis

☐ Deep back

STRUCTURES TO IDENTIFY

- ☐ Splenius capitis
- ☐ Splenius cervicis
- ☐ Semispinalis
- ☐ Multifidus
- ☐ Quadratus lumborum
- ☐ Rotatores
- ☐ Interspinales
- ☐ Intertransversarii

☐ Thoracic wall

Be prepared to state the fiber direction of the external and internal intercostals and what each does to the ribs.

STRUCTURES TO IDENTIFY

- ☐ External intercostals
- ☐ Internal intercostals
- ☐ Innermost intercostals
- ☐ Transversus thoracis
- ☐ Serratus posterior superior
- ☐ Serratus posterior inferior
- ☐ Intercostal space
- ☐ Neurovascular bundle in the costal groove

☐ The diaphragm

Name what passes through each of the three openings.

STRUCTURES TO IDENTIFY

- ☐ Central tendon
- ☐ Right crus
- ☐ Left crus
- ☐ Costal part
- ☐ Sternal part
- ☐ Caval opening (T8)
- ☐ Esophageal hiatus (T10)
- ☐ Aortic hiatus (T12)
- ☐ Phrenic nerve

☐ Anterior thorax

STRUCTURES TO IDENTIFY

- ☐ Pectoralis major
- ☐ Pectoralis minor
- ☐ Serratus anterior
- ☐ Subclavius
- ☐ External oblique, upper slips
- ☐ Clavipectoral fascia

Which nerve supplies serratus anterior, and what deficit follows its injury?

____

CHAPTER 3

Muscles of the Upper Extremity

SHOULDER, ARM, FOREARM, HAND

Learn these by compartment. On a cadaver or a model the compartment tells you the action and the nerve before you have identified the individual muscle.

☐ Rotator cuff

Name the tubercle each one reaches and the movement each one produces.

STRUCTURES TO IDENTIFY

- ☐ Supraspinatus
- ☐ Infraspinatus
- ☐ Teres minor
- ☐ Subscapularis

☐ Scapulohumeral, other

STRUCTURES TO IDENTIFY

- ☐ Deltoid, anterior fibers
- ☐ Deltoid, middle fibers
- ☐ Deltoid, posterior fibers
- ☐ Teres major
- ☐ Coracobrachialis

☐ Arm, anterior compartment

STRUCTURES TO IDENTIFY

- ☐ Biceps brachii, long head
- ☐ Biceps brachii, short head
- ☐ Bicipital aponeurosis
- ☐ Brachialis
- ☐ Coracobrachialis

☐ Arm, posterior compartment

STRUCTURES TO IDENTIFY

- ☐ Triceps brachii, long head
- ☐ Triceps brachii, lateral head
- ☐ Triceps brachii, medial head
- ☐ Anconeus
- ☐ Radial groove

☐ Forearm, anterior compartment

STRUCTURES TO IDENTIFY

- ☐ Brachioradialis
- ☐ Pronator teres
- ☐ Flexor carpi radialis
- ☐ Palmaris longus
- ☐ Flexor carpi ulnaris
- ☐ Flexor digitorum superficialis
- ☐ Flexor digitorum profundus
- ☐ Flexor pollicis longus
- ☐ Pronator quadratus
- ☐ Medial epicondyle, common flexor origin

☐ Forearm, posterior compartment

STRUCTURES TO IDENTIFY

- ☐ Extensor carpi radialis longus
- ☐ Extensor carpi radialis brevis
- ☐ Extensor digitorum
- ☐ Extensor digiti minimi
- ☐ Extensor carpi ulnaris
- ☐ Supinator
- ☐ Abductor pollicis longus
- ☐ Extensor pollicis brevis
- ☐ Extensor pollicis longus
- ☐ Extensor indicis
- ☐ Lateral epicondyle, common extensor origin

☐ Hand, intrinsic muscles

STRUCTURES TO IDENTIFY

- ☐ Thenar eminence
- ☐ Abductor pollicis brevis
- ☐ Flexor pollicis brevis
- ☐ Opponens pollicis
- ☐ Adductor pollicis
- ☐ Hypothenar eminence
- ☐ Abductor digiti minimi
- ☐ Flexor digiti minimi brevis
- ☐ Opponens digiti minimi
- ☐ Lumbricals
- ☐ Palmar interossei
- ☐ Dorsal interossei

☐ Landmarks and retinacula

STRUCTURES TO IDENTIFY

- ☐ Cubital fossa
- ☐ Carpal tunnel
- ☐ Flexor retinaculum
- ☐ Extensor retinaculum
- ☐ Palmar aponeurosis
- ☐ Anatomical snuffbox
- ☐ Extensor expansion

Name the two common origins that most superficial forearm muscles share.

____

CHAPTER 4

The Heart

MODEL, CADAVER, AND SLIDE

Work the heart from the outside in: sac, wall, chambers, valves, then the vessels on its surface. Be prepared to trace blood through it in order on any specimen you are given.

☐ Position and coverings

STRUCTURES TO IDENTIFY

- ☐ Mediastinum
- ☐ Apex
- ☐ Base
- ☐ Anterior (sternocostal) surface
- ☐ Inferior (diaphragmatic) surface
- ☐ Fibrous pericardium
- ☐ Parietal serous pericardium
- ☐ Visceral serous pericardium (epicardium)
- ☐ Pericardial cavity

☐ The heart wall

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Epicardium
- ☐ Myocardium
- ☐ Endocardium

☐ External surface features

STRUCTURES TO IDENTIFY

- ☐ Coronary sulcus
- ☐ Anterior interventricular sulcus
- ☐ Posterior interventricular sulcus
- ☐ Right auricle
- ☐ Left auricle
- ☐ Ligamentum arteriosum

☐ Chambers and internal features

STRUCTURES TO IDENTIFY

- ☐ Right atrium
- ☐ Right ventricle
- ☐ Left atrium
- ☐ Left ventricle
- ☐ Interatrial septum
- ☐ Interventricular septum
- ☐ Fossa ovalis
- ☐ Pectinate muscles
- ☐ Crista terminalis
- ☐ Sinus venarum
- ☐ Trabeculae carneae
- ☐ Papillary muscles
- ☐ Chordae tendineae
- ☐ Moderator band
- ☐ Conus arteriosus
- ☐ Aortic vestibule

☐ Valves

For each valve, name what it separates and what it prevents.

STRUCTURES TO IDENTIFY

- ☐ Tricuspid valve
- ☐ Mitral (bicuspid) valve
- ☐ Pulmonary semilunar valve
- ☐ Aortic semilunar valve
- ☐ Fibrous skeleton

☐ Great vessels

STRUCTURES TO IDENTIFY

- ☐ Superior vena cava
- ☐ Inferior vena cava
- ☐ Pulmonary trunk
- ☐ Right pulmonary artery
- ☐ Left pulmonary artery
- ☐ Pulmonary veins
- ☐ Ascending aorta
- ☐ Aortic arch
- ☐ Descending thoracic aorta

☐ Coronary circulation

STRUCTURES TO IDENTIFY

- ☐ Right coronary artery
- ☐ Right marginal artery
- ☐ Posterior interventricular artery (PDA)
- ☐ Left coronary artery
- ☐ Anterior interventricular artery (LAD)
- ☐ Circumflex artery
- ☐ Left marginal artery
- ☐ Coronary sinus
- ☐ Great cardiac vein
- ☐ Middle cardiac vein
- ☐ Small cardiac vein

☐ Conduction system, on the model

STRUCTURES TO IDENTIFY

- ☐ SA node
- ☐ AV node
- ☐ AV bundle (bundle of His)
- ☐ Right bundle branch
- ☐ Left bundle branch
- ☐ Purkinje fibers

☐ Cardiac muscle slide

FUNCTION ____________________ LOCATION ____________________

SPECIAL STRUCTURES ____________________

STRUCTURES TO IDENTIFY

- ☐ Cardiomyocyte
- ☐ Central nucleus
- ☐ Branching fibers
- ☐ Intercalated discs
- ☐ Striations

Trace a drop of blood from the superior vena cava to the aorta, naming every chamber and valve.

____

____

CHAPTER 5

Blood Vessels, Histology

SLIDES AND WALL MODELS

Almost every question here is answered from the wall. Identify the tunics first, then use their proportions to name the vessel.

☐ Artery, cross section

FUNCTION ____________________ LOCATION ____________________

SPECIAL STRUCTURES ____________________

STRUCTURES TO IDENTIFY

- ☐ Tunica intima
- ☐ Endothelium
- ☐ Basement membrane
- ☐ Internal elastic lamina
- ☐ Tunica media
- ☐ Smooth muscle
- ☐ External elastic lamina
- ☐ Tunica externa
- ☐ Vasa vasorum
- ☐ Lumen

☐ Vein, cross section

FUNCTION ____________________ LOCATION ____________________

SPECIAL STRUCTURES ____________________

STRUCTURES TO IDENTIFY

- ☐ Tunica intima
- ☐ Tunica media
- ☐ Tunica externa
- ☐ Venous valve
- ☐ Lumen

Name the two features you would use first to tell an artery from a vein on a slide.

____

☐ Elastic artery

Name two elastic arteries you would expect to see labelled.

STRUCTURES TO IDENTIFY

- ☐ Elastic lamellae
- ☐ Thick tunica media
- ☐ Internal elastic lamina
- ☐ External elastic lamina

☐ Muscular artery

STRUCTURES TO IDENTIFY

- ☐ Prominent internal elastic lamina
- ☐ Smooth muscle of the tunica media
- ☐ Thin external elastic lamina

☐ Arteriole and capillary bed

STRUCTURES TO IDENTIFY

- ☐ Arteriole
- ☐ Metarteriole
- ☐ Precapillary sphincter
- ☐ Capillary
- ☐ Capillary bed
- ☐ Thoroughfare channel
- ☐ Postcapillary venule
- ☐ Muscular venule

☐ Capillary types

For each type, name one organ where it is found and state what can cross its wall.

STRUCTURES TO IDENTIFY

- ☐ Continuous capillary
- ☐ Fenestrated capillary
- ☐ Sinusoid
- ☐ Intercellular cleft
- ☐ Fenestration
- ☐ Basement membrane

☐ Fetal circulation, on the model

Name the adult remnant of each fetal structure.

STRUCTURES TO IDENTIFY

- ☐ Placenta
- ☐ Umbilical vein
- ☐ Ductus venosus
- ☐ Foramen ovale
- ☐ Ductus arteriosus
- ☐ Umbilical arteries

CHAPTER 6

Arteries and Veins of the Thorax and Upper Limb

NAMED VESSELS, CADAVER AND MODEL

These are routes. Trace each one continuously with your finger before you try to name segments, and learn the landmark where each name changes.

☐ Aortic arch and its branches

Be prepared to explain why the right and left sides are not symmetrical.

STRUCTURES TO IDENTIFY

- ☐ Ascending aorta
- ☐ Aortic arch
- ☐ Brachiocephalic trunk
- ☐ Right common carotid artery
- ☐ Right subclavian artery
- ☐ Left common carotid artery
- ☐ Left subclavian artery
- ☐ Descending thoracic aorta

☐ Other thoracic arteries

STRUCTURES TO IDENTIFY

- ☐ Internal thoracic artery
- ☐ Anterior intercostal arteries
- ☐ Posterior intercostal arteries
- ☐ Bronchial arteries
- ☐ Esophageal arteries
- ☐ Vertebral artery
- ☐ Thyrocervical trunk
- ☐ Costocervical trunk

☐ Arteries of the upper limb

Name the landmark at which the subclavian becomes the axillary, and the axillary becomes the brachial.

STRUCTURES TO IDENTIFY

- ☐ Subclavian artery
- ☐ Axillary artery
- ☐ Thoracoacromial artery
- ☐ Lateral thoracic artery
- ☐ Subscapular artery
- ☐ Anterior circumflex humeral artery
- ☐ Posterior circumflex humeral artery
- ☐ Brachial artery
- ☐ Deep brachial artery
- ☐ Radial artery
- ☐ Ulnar artery
- ☐ Common interosseous artery
- ☐ Superficial palmar arch
- ☐ Deep palmar arch
- ☐ Digital arteries

☐ Superficial veins of the upper limb

STRUCTURES TO IDENTIFY

- ☐ Cephalic vein
- ☐ Basilic vein
- ☐ Median cubital vein
- ☐ Median antebrachial vein
- ☐ Dorsal venous network of the hand

☐ Deep veins of the thorax and upper limb

STRUCTURES TO IDENTIFY

- ☐ Radial veins
- ☐ Ulnar veins
- ☐ Brachial veins
- ☐ Axillary vein
- ☐ Subclavian vein
- ☐ Internal jugular vein
- ☐ Brachiocephalic veins
- ☐ Superior vena cava
- ☐ Azygos vein
- ☐ Hemiazygos vein
- ☐ Internal thoracic vein

Why is there one brachiocephalic artery but two brachiocephalic veins?

CHAPTER 7

Nerves of the Upper Limb

THE BRACHIAL PLEXUS

Build the plexus by level before you learn individual nerves. The counts, five roots, three trunks, six divisions, three cords, five branches, are how you check your own work.

☐ Levels of the plexus

The cords are named for their position around the axillary artery. Identify the artery first.

STRUCTURES TO IDENTIFY

- ☐ Roots, C5 to T1
- ☐ Upper trunk
- ☐ Middle trunk
- ☐ Lower trunk
- ☐ Anterior divisions
- ☐ Posterior divisions
- ☐ Lateral cord
- ☐ Posterior cord
- ☐ Medial cord

☐ Terminal branches

For each nerve, name the compartment it supplies and the deficit its injury produces.

STRUCTURES TO IDENTIFY

- ☐ Musculocutaneous nerve
- ☐ Median nerve
- ☐ Ulnar nerve
- ☐ Radial nerve
- ☐ Axillary nerve

☐ Branches above the terminal nerves

STRUCTURES TO IDENTIFY

- ☐ Dorsal scapular nerve
- ☐ Long thoracic nerve
- ☐ Suprascapular nerve
- ☐ Nerve to subclavius
- ☐ Lateral pectoral nerve
- ☐ Medial pectoral nerve
- ☐ Thoracodorsal nerve
- ☐ Upper subscapular nerve
- ☐ Lower subscapular nerve
- ☐ Medial cutaneous nerve of the arm
- ☐ Medial cutaneous nerve of the forearm

☐ Nerves at their vulnerable points

Name the injury associated with each site and the sign it produces.

STRUCTURES TO IDENTIFY

- ☐ Axillary nerve at the surgical neck of the humerus
- ☐ Radial nerve in the radial groove
- ☐ Ulnar nerve behind the medial epicondyle
- ☐ Median nerve in the carpal tunnel
- ☐ Long thoracic nerve on the chest wall

☐ Thoracic nerves

STRUCTURES TO IDENTIFY

- ☐ Intercostal nerves
- ☐ Phrenic nerve
- ☐ Vagus nerve in the thorax
- ☐ Sympathetic trunk

CHAPTER 8

Blood

THE SMEAR

One slide, and every question comes off the nucleus. Identify each cell by the shape of its nucleus first and the granules second.

☐ Whole blood, in a spun tube

STRUCTURES TO IDENTIFY

- ☐ Plasma layer
- ☐ Buffy coat
- ☐ Erythrocyte layer
- ☐ Hematocrit

☐ Erythrocytes

FUNCTION ______________________ LOCATION ______________________

SPECIAL STRUCTURES ______________________

STRUCTURES TO IDENTIFY

- ☐ Biconcave disc
- ☐ Pale central area
- ☐ Absence of a nucleus
- ☐ Rouleaux formation

☐ Granulocytes

STRUCTURES TO IDENTIFY

- ☐ Neutrophil
- ☐ Eosinophil
- ☐ Basophil
- ☐ Multi-lobed nucleus
- ☐ Bilobed nucleus
- ☐ Nucleus obscured by dark granules
- ☐ Fine pale granules
- ☐ Coarse red-orange granules

☐ Agranulocytes

STRUCTURES TO IDENTIFY

- ☐ Lymphocyte
- ☐ Monocyte
- ☐ Abundant pale cytoplasm
- ☐ Large round nucleus with thin rim of cytoplasm
- ☐ Kidney-shaped nucleus

☐ Platelets and formation

STRUCTURES TO IDENTIFY

- ☐ Platelet (thrombocyte)
- ☐ Red bone marrow
- ☐ Reticulocyte
- ☐ Megakaryocyte
- ☐ Hemocytoblast

Order the five leukocytes from most to least numerous.